

SDS 4135 - Project 2

Due September 27

Grand Blvd Bus Data Exploratory Analysis

Joseph Silagi from St. Louis Metro Transit has provided us with one year of data from the #70 Grand bus line. Each row of the data contains information from one stop from one trip of a bus driving the route, and there are rows for both the northbound (`route_direction_id = 7`) and the southbound (`route_direction_id = 9`) routes. Each row contains information about the time of day (in seconds after midnight) that the bus arrived, how early or late it was (adherence) how many people boarded, how many people exited (alightings), and various other pieces of information, such as which physical bus the data came from.

Please note and abide by the data usage statement from Metro Transit:

“The accompanying dataset (`70_grand_20250701-20260630.csv`) is provided exclusively for academic instruction. It is not official Metro Transit/Bi-State Development data, nor does it reflect the agency’s system of record. Students may use this dataset only within the scope of this course. Redistribution or external use is not permitted.”

I have done some initial cleaning of this dataset, mostly to aggregate information from duplicate rows. You have access to a derived dataset `bus_stop_data_v1.csv`. The data usage statement of course applies to any derived dataset as well.

Your overall task is to get familiar with the data by providing exploratory analyses and data visualizations. Each group must investigate the topic assigned to their group number below:

1. Variation in number of boardings and number of passengers by location, time of day, day of week, etc.
2. Variation in route duration.
3. Variation average adherence along the route.
4. Variation in stop time across time of day, day of week, number of boardings
5. Variation in travel time between stops
6. Identify potentially erroneous outliers in the data
7. Identify reasons for missing values in the data

In addition to your assigned topic, each group must pick one additional topic to investigate, either from the list above, or a topic you come up with on your own.

Your report should also outline important data cleaning and processing steps.